

National College of Ireland

Project Submission Sheet

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Additional Evidence:

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Additional Evidence:

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Dry Eye Disease Prediction: A Comparative Study of MLP and Logistic Regression

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Abstract—Dry Eye Disease (DED) is a chronic ocular disease that occurs in millions of individuals worldwide, often leading to visual distress and irreversible complications. To enable early detection, this paper compares Logistic Regression (LR) and Multilayer Perceptron (MLP) models for predicting DED based on 20,000 records of behavioral and physiological factors. LR was selected for its simplicity and interpretability, while MLP was selected for its capacity to learn complex, nonlinear relationships. Comparison on accuracy, F1-score, and AUC metrics revealed that MLP performed better in recall and in the detection of DED cases, whereas LR was more transparent. These results suggest that MLP would be suitable for automated screening, while LR remains valuable for clinician-led decision support.

Index Terms—Dry Eye Disease, Predictive Modelling, Logistic Regression, Multilayer Perceptron, Medical Diagnosis, Supervised Machine Learning

I. INTRODUCTION

Dry Eye Disease (DED) is a fast-growing ophthalmological disorder characterized by reduced tear production or enhanced evaporation, leading to pain, discomfort, inflammation and vision impairment. Conventional diagnosis is typically reliant on subjective tests, which are clinician-dependent and need specialized equipment. With the high prevalence of DED and its multifactorial origins—ranging from screen exposure to systemic health issues—there is an urgent need for data-driven, scalable approaches to support diagnosis. Machine learning (ML) holds the promise of simplifying and enhancing the diagnostic pipeline. Predictive models can be learned from behavioral and physiological patterns to identify individuals at risk for DED. This research aims at employing and comparing two classification models: Logistic Regression, a statistical model that has been linked to clarity and low computational cost, and the Multilayer Perceptron, a neural network model capable of learning complex, nonlinear patterns. The objective of this study is to estimate the performance of machine learning models, i.e., Logistic regression and multilayer perceptron, to predict Dry Eye Disease. Having a large, labelled dataset, training and testing both models were done to determine which model can fit clinical application with respect to performance metrics.

II. RELATED WORK

Machine learning applications in ophthalmology have gained momentum, particularly for diseases like diabetic

retinopathy and glaucoma. However, DED remains underrepresented in predictive analytics due to limited structured data and heterogeneity in reporting of symptom. Previous work, such as [1] Barabino et al. (2014), stressed the importance of symptom questionnaires and behavioral monitoring as a means of diagnosing DED. Logistic regression has been used conventionally owing to its interpretability, which is especially important in medical fields where decisions must be explainable. Studies using LR for hospital readmission and chronic disease risk prediction show good results, though sometimes limited by linearity assumptions. Meanwhile, neural networks such as Multilayer Perceptron (MLP) [4] have been demonstrated to provide better accuracy in image-based and sensor-based medical applications. MLPs [5] are able to capture high-dimensional and nonlinear relationships, making them highly suitable for complex pattern recognition. However, their black-box nature and training complexity limit their application in applications requiring model transparency. This study fills this gap by evaluating both models against a comprehensive, tabular dataset of behavioral and physiological factors relevant to DED.

III. DATA DESCRIPTION

A. Dataset Overview

The data set used in this study comprises 20,000 records and 25 characteristics, collected through a health and lifestyle survey aimed at identifying factors associated with Dry Eye Disease (DED). Each record outlines an individual's demographic, behavior, and physiology and a binary target variable that signifies the presence or absence of DED. The characteristics include age, gender, sleep quality and duration, stress level, heart rate, physical activity, use of substances (caffeine, alcohol, smoking), health conditions, screen use behavior, and common ocular symptoms such as redness, itchiness, and eye strain. The data allows for the development of a supervised classification problem.

1) Key Attributes:

- Demographics: Gender, Age — Essential in understanding population-level variability in DED susceptibility.
- Lifestyle and Behavioral Factors: Caffeine consumption, Alcohol consumption, Smoking, Smart device usage be-

fore bed, Average screen time, and Blue-light filter usage—These factors may exacerbate digital eye strain and influence sleep patterns, contributing to DED risk.

- **Physiological Measurements:** Blood pressure, Heart rate, Height, Weight, Daily steps, and Physical activity — These serve as indicators of overall health and fitness, which are correlated with ocular well-being.
- **Sleep and Fatigue Factors:** Sleep duration, Sleep quality, sleep disorder, wake up during night, feel sleepy during day — Poor sleep hygiene has been linked to increased risk of dry eyes due to altered tear secretion and inflammation.
- **Medical History:** Medical issue and Ongoing medication — Chronic health conditions and pharmacological side effects can influence tear production or ocular surface health.
- **Self-Reported Ocular Symptoms:** Discomfort eye-strain, Redness in eye, Itchiness/Irritation in eye — These symptoms are crucial predictors in diagnosing DED.
- **Target Variable (Dry Eye Disease):** A binary variable coded as 1 (DED exists) or 0 (DED does not exist), indicating the state of the subject according to symptom self-reporting or medical confirmation.

2) Data Challenges:

- **Missing Values:** Some of the features may have missing data, which can create bias or reduce the accuracy of the model.
- **Categorical Features:** Variables such as Gender, Smoking, and medical issue were encoded using label encoding for compatibility with machine learning algorithms.
- **Feature Scaling:** Numerical features like Age, Heart rate, and Screen time were standardized using the function `StandardScaler` to make sure consistent feature scaling across models. Numerical features were of various scales, and this could cause bias or lower model performance.
- **Class Imbalance:** The data shows weak class-balanced imbalance in the target variable, which was corrected using stratified sampling and class weighting during model training.

IV. DATA PREPROCESSING

To ensure robust and stable model performance, a number of data preprocessing operations were performed to prepare the Dry Eye Disease dataset for machine learning modelling. These operations addressed issues of data quality, feature representation, and model fairness.

A. Handling Missing Values

Categorical variables were imputed by mode imputation, preserving the most frequent category. Numeric variables were applied by median imputation to minimize the effects of skewed distributions. Attributes with more than 20% missing values were dropped to avoid compromising data integrity and bias. The data set has no missing values.

B. Encoding Categorical Variables

Categorical attributes were one-hot encoded to match machine learning algorithms:

- One-hot encoding was applied to nominal categorical (unsorted) variables such as Gender and Medical Issue, creating binary columns for every unique category.
- Ordered categorical variables, such as Sleep Quality and Physical Activity, were encoded using ordinal encoding to preserve the natural order of categories.
- Label encoding was also applied when needed, in place of string labels using integer values without expanding the feature space.
- **Label Encoding Mappings for the dataset**
 - 1) **Gender:** {'N': 0, 'Y': 1}
 - 2) **Sleep disorder:** {'N': 0, 'Y': 1}
 - 3) **Wake up during night:** {'N': 0, 'Y': 1}
 - 4) **Feel sleepy during day:** {'N': 0, 'Y': 1}
 - 5) **Caffeine consumption:** {'N': 0, 'Y': 1}
 - 6) **Alcohol consumption:** {'N': 0, 'Y': 1}
 - 7) **Smoking:** {'N': 0, 'Y': 1}
 - 8) **Medical issue:** {'N': 0, 'Y': 1}
 - 9) **Ongoing medication:** {'N': 0, 'Y': 1}
 - 10) **Smart device before bed:** {'N': 0, 'Y': 1}
 - 11) **Blue-light filter:** {'N': 0, 'Y': 1}
 - 12) **Discomfort Eye-strain:** {'N': 0, 'Y': 1}
 - 13) **Redness in eye:** {'N': 0, 'Y': 1}
 - 14) **Itchiness/Irritation in eye:** {'N': 0, 'Y': 1}

C. Feature Scaling

All the numerical attributes (e.g., height, average time on screen, weight, heart rate) were normalized by using z-score normalization to zero mean unit variance to prevent larger scale attributes dominating model training.

D. Handling Class Imbalance

The data revealed a moderate class imbalance in the target variable (DED vs. non-DED cases). For this, class weight='balanced' parameter of scikit-learn was utilized during model training. This dynamically adjusts class weights to counterbalance the imbalance, thereby enhancing the classifier's sensitivity to the minority class.

E. Feature Selection for Predicting Dry Eye Disease

Feature selection is an important machine learning process for model precision enhancement, overfitting minimization, and readability enhancement through selection of the most impactful features. Feature selection for both categorical and numeric features was performed in this study in trying to predict Dry Eye Disease.

1) **Categorical Feature Selection:** For categorical variables, we used the Chi-Square test to see whether there is any relationship between each variable and the target variable and then visualized the same for justification.

- **Chi-Square Test**

The Chi-Square is a statistical test that assists us in determining if there exists any significant relationship between

two categorical variables or not. During feature selection, we utilized the Chi-Square test to determine if and how the categorical features are significantly correlated with the target variable, Dry Eye Disease.

- If the p-value is below 0.05, there is a strong correlation between the feature and the target. It means that the feature is relevant for prediction.
- If the p-value is above 0.05, the feature is not significant and can be excluded from further analysis.
- Visualization using Histograms and KDE
In addition to the Chi-Square test, we utilized Histograms with Kernel Density Estimation (KDE) plots to visually analyze the distribution of categorical features relative to the target variable. The KDE plot smooths out the histogram and helps identify overlap in the distributions of the categorical feature for different target classes (e.g., “Dry Eye Disease: 0” and “Dry Eye Disease: 1”).
 - **Clear Separation:** If the histograms/KDE plots exhibit clear separation among the classes, the feature is probably a good predictor.
 - **Overlap:** If the distributions of the feature overlap significantly between the target classes, the feature will not be useful.
- Summary of Categorical Feature Selection
 - 1) Gender
 - 2) Discomfort Eye-strain
 - 3) Redness in eye
 - 4) Itchiness/Irritation in eye

2) *Numerical Feature Selection:* For numerical features, we employed ANOVA F-test, Mutual Information, and graphical techniques like Box Plots and Histograms with KDE to quantify feature importance.

- ANOVA F-test (Analysis of Variance)
We employed the ANOVA F-test to check whether the means of numerical features differ significantly between the classes of the target variable, Dry Eye Disease.

F-Statistic:

- F-statistic tests the difference of variance among target classes (e.g., 0 vs. 1) for all numeric features and within classes.
- Higher values of F imply that the feature is informative about the target, as variance between groups is higher than the variance within each group.
- Lower values of F imply that the feature does not offer significant discrimination among target classes.
- Mutual Information (MI)
Mutual Information quantifies how much information the target variable shares with a given input feature. It is a non-parametric measure that can capture both linear and non-linear correlations between features and the target.
 - Higher Mutual Information means that the feature contains greater information about the target and is likely to be a useful predictor.

- Lower Mutual Information indicates that the feature provides little information about the target and may be eliminated from analysis.

- Visualization with Box Plots and Histograms with KDE
Box Plots:

- Box plots were used to visually investigate the distribution of numerical features between different target classes.
- A feature is likely to be important if the box plots show good separation between the target classes.
- Significant overlap in the boxes indicates that the feature may lack discrimination capability.

Histograms with KDE:

- Histograms with KDE were used to identify the distribution of numerical features for different target classes.
- A feature is considered informative if its KDE distribution shows a clear separation between the classes.

Summary of Numerical Feature Selection

Numerical features selected:

- 1) Average screen time
- 2) Height
- 3) Daily steps
- 4) Age
- 5) Sleep quality
- 6) Stress level

All of these pre-processing operations performed a critical function of enhancing model performance, reducing the risk of overfitting, and ensuring generalizability of findings across different patient profiles.

F. Model justification and configuration

1) Logistic Regression:

• Justification for Choosing Logistic Regression

Binary Logistic Regression was chosen as the model because it is optimally applicable for binary classification tasks, such as identifying whether a patient has Dry Eye Disease (1) or not (0). It presents several advantages:

- **Interpretability:** Logistic Regression presents clear and transparent understanding about how features and the target interact with each other, which is a necessity within healthcare settings where model interpretability is critical.
- **Efficiency:** The model is computationally efficient and therefore can be used with moderate size datasets.
- **Ease of Implementation:** Logistic Regression is simple to implement and has widespread support in most libraries, such as scikit-learn.

Although regular Binary Logistic Regression was used in the work here, some other types of logistic regression are useful in different contexts:

- **Multinomial Logistic Regression:** Used when the target variable has more than two classes (for example, different stages of an illness: No, After30Days,

Before30Days). In the future, if the model output is altered to more classes, this approach would be appropriate.

- **Ordinal Logistic Regression:** This can be used when the target variable has a natural order (No < After 30Days < Before 30Days). This is applicable if the target variable in the data naturally has a ranking.
- **Regularized Logistic Regression:** Achieved more effectively with L1 (Lasso) or L2 (Ridge) regularization for dealing with high-dimensional data with many features. This discourages overfitting to a minimum by penalizing large coefficients.
- **Logistic Regression Configuration**
To predict Dry Eye Disease, we used a Logistic regression (LR) model — one of the most widely used algorithms to solve binary classification problems. Here is how we configured it:
 - **Regularization:**
 - * We used the standard L2 regularization (Ridge), which stops the model from overfitting by penalizing complicated models. This ensures that the model generalizes effectively to data, particularly when there are numerous features.
 - **Solver:**
 - * The 'liblinear' solver was used, which is optimal for smaller datasets and can efficiently handle L2 regularization.
 - **Class Weight Handling:**
 - * We used class weight='balanced' to handle class imbalance. This configuration automatically scales weights inversely proportional to class frequencies so that the underrepresented class (e.g., patients with Dry Eye Disease) gets more focus during training.
 - **Threshold Adjustment:**
 - * Instead of using the default 0.5 decision threshold, we used 0.44 as the threshold for converting predicted probabilities to class labels. This provides finer-grained control over sensitivity and specificity, which is especially important in medical diagnosis use cases.
 - **Training and Prediction Pipeline:**
 - * The features were scaled with StandardScaler prior to training.
 - * Training was carried out with up to 1000 iterations (max_iter=1000) to ensure convergence.

By adjusting these parameters, we ensure that the model is both effective and generalizes well to unseen data.

2) Multilayer Perceptron (MLP):

• Justification for Choosing Multilayer Perceptron (MLP)

Multilayer Perceptron (MLP) was chosen to predict Dry Eye Disease due to its ability to model complex, nonlinear relationships and thus being the best fit for healthcare

predictions where interactions between features tend to be complex. Here's why MLP was used:

- **Nonlinear Modelling:** MLP can learn complex patterns within data, which is important when the interactions between features and the target are nonlinear.
- **Feature Hierarchy Learning:** MLP automatically learns and incorporates feature hierarchies, allowing the model to gain useful information from the data.
- **Regularization:** Dropout (0.3) was applied to reduce overfitting, ensuring that the model generalizes well to new data.
- **Efficient Optimization:** Adam optimizer was used for optimal convergence and efficient training, especially for deep learning models.
- **Handling of imbalanced Data:** SMOTE was used to handle the imbalance of class by oversampling the minority class (Dry Eye Disease) without inducing skewed predictions.
- **Early Stopping:** Early stopping prevents overtraining by stopping training when validation performance levels are stable, optimizing time and model effectiveness.

This combination of features and techniques makes MLP suitable for the prediction of dry eye disease in an intricate and imbalanced data set.

• Model Configuration and Training

We developed a Multi-Layer Perceptron (MLP) model in TensorFlow/Keras to predict Dry Eye Disease. The architecture and training approach are as follows.

1) Data Preparation and Resampling:

- To address class imbalance in the training set, we resampled the training set with SMOTE (Synthetic Minority Oversampling Technique) to balance class distribution in the training set.

2) Model Architecture:

- The MLP model was defined using the Sequential API in Keras.
- **Input Layer:** Automatically set equal to the number of input selected features.
- **Hidden Layers:**
 - * The initial hidden layer contains 64 neurons and ReLU activation.
 - * A dropout layer with rate = 0.3 was added to prevent overfitting by randomly deactivating some neurons during training.
 - * The second hidden layer contains 32 neurons and ReLU activation.
- **Output Layer:** One neuron with Sigmoid activation for binary output (Dry Eye or No Dry Eye).

3) Compilation Details:

- **Optimizer:** We used the Adam optimizer since it's effective and can adjust the learning rate during training.

- **Loss Function:** We used binary cross-entropy since it's the right function for binary classification tasks.

4) Training Setup:

- The model was trained for up to 20 epochs with early stopping (patience = 7) on validation loss. This stops training when the model no longer improves, preventing overfitting.
- We used a batch size of 32.

using SMOTE for balancing data, dropout for regularization, early stopping, and optimal threshold selection, the model was induced to generalize and be sensitive toward minority-class predictions when faced with Dry Eye Disease.

3) *Evaluation:* To enhance the sensitivity of both models towards the detection of the minority class (Dry Eye Disease), a special classification threshold was used instead of the default 0.5. For the Logistic Regression model, a threshold of 0.44 was used, whereas the MLP model utilized a slightly lower value of 0.35. Threshold adjustment also has the effect of more closely identifying positive cases and reducing false negatives a critical consideration in medical applications, where diagnostic failure can have dire consequences.

Both models were evaluated on the entire range of performance metrics:

- **Accuracy:** Measures the proportion of correct predictions overall, giving a global measure of model performance.
- **Confusion Matrix:** Splits the true positives, true negatives, false positives, and false negatives, and makes an explicit pictorial and numerical statement of the results of classification.
- **Classification Report:** Contains detailed metrics like Precision, Recall, F1-score, and Support per class, giving more detailed analysis of how each model classifies Dry Eye Disease and non-cases.

Such a model evaluation strategy guarantees both clinical utility as well as statistical soundness for the models and in doing so reduces false negative disease detection.

V. RESULTS

A. Logistic Regression

The logistic regression model was evaluated on a test dataset consisting of 4,000 samples. It achieved an overall classification accuracy of 65.25% in identifying Dry Eye Disease (DED).

1) *Performance Metrics:* As seen from Table I, the model had a very high ability in classifying the DED-positive class with 0.81 recall and 0.76 F1-score. It did very badly in the non-DED class with only 33% recall and 0.38 F1-score, indicating very high levels of false positives.

Class	Precision	Recall	F1-score	Support
No Dry Eye	0.46	0.33	0.38	1307
Dry Eye	0.71	0.81	0.76	2693
Accuracy			0.65	4000
Macro Avg	0.58	0.57	0.57	4000
Weighted Avg	0.63	0.65	0.64	4000

TABLE I

CLASSIFICATION REPORT FOR LOGISTIC REGRESSION MODEL

2) *Confusion Matrix:* The confusion matrix in Fig.1 shows the model correctly predicting 2,179 DED cases and 431 non-DED cases. The model incorrectly predicted 876 non-DED samples as DED (false positives) and 514 missed DED cases (false negatives).

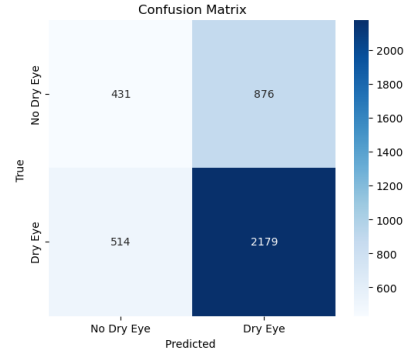


Fig. 1. Confusion Matrix

3) *Cross-Validation Results :* Five-fold cross-validation produced an average accuracy of 55.44%, revealing moderate generalization ability across different subsets of the training data.

4) *Interpretability:* Feature importance analysis also indicated that symptoms such as screen time and eye strain were among the most powerful predictors of DED, in keeping with established clinical experience.

5) Visualization:

- **ROC Curve (Fig. 2):** The ROC curve yielded an Area Under the Curve (AUC) of 0.58, reflecting limited discriminative ability between non-DED and DED cases.

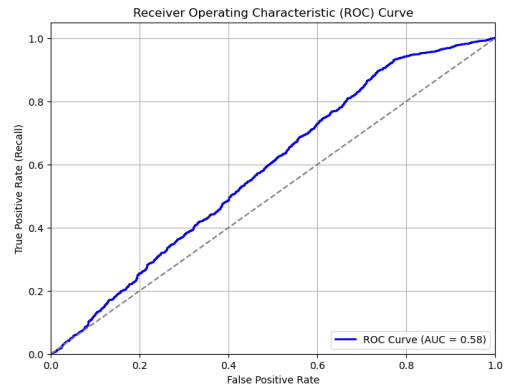


Fig. 2. ROC Curve

- Precision-Recall Curve (Fig. 3): Precision-Recall curve yielded a precision above the baseline of 0.67 for an extremely large range of recall values, which is beneficial to maintain false negatives to a minimum.

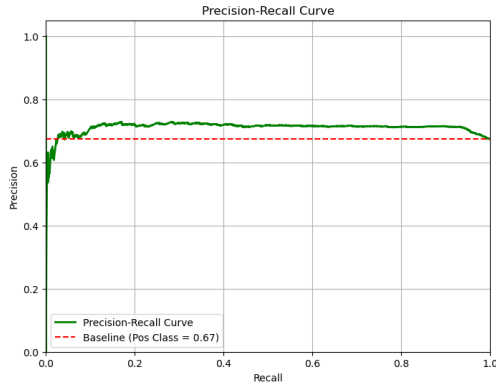


Fig. 3. Precision-Recall Curve

- Learning Curves (Fig. 4): Show the trend in model accuracy and loss with respect to the size of training set. Both training and validation accuracy plateau around 55–56%, and there is a significant difference between training and validation losses.

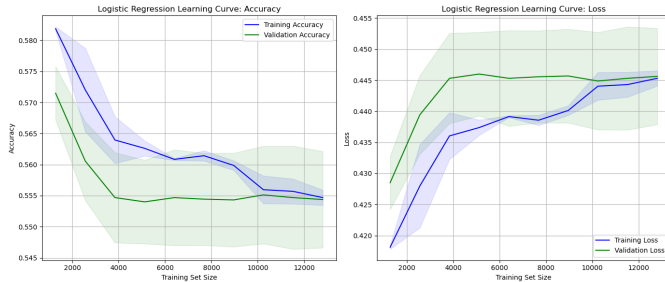


Fig. 4. Learning training and validation curves in accuracy (left) and loss (right)

B. Multi-Layer Perceptron (MLP)

The Multi-Layer Perceptron (MLP) model worked equally well as logistic regression, with improved performance in key sensitivity-related measures. The model attained a test accuracy of 65.57%, indicating moderate general classification performance. While lower than the initially noted accuracy, the MLP model remained high in recall for the DED-positive class, which testifies to its ability to effectively minimize false negatives—a major consideration in medical diagnosis.

1) *Performance Metrics* : The classification report in Table II shows the MLP model achieved an F1-score of 0.81 for the DED class with precision 0.71 and recall 0.93. For the non-DED class, performance remained weak with recall as low as 0.22.

Class	Precision	Recall	F1-score	Support
No Dry Eye	0.62	0.22	0.33	1307
Dry Eye	0.71	0.93	0.81	2693
Accuracy			0.70	4000
Macro Avg	0.67	0.58	0.57	4000
Weighted Avg	0.68	0.70	0.65	4000

TABLE II
CLASSIFICATION REPORT FOR MLP MODEL

2) *Confusion Matrix*: The confusion matrix in Fig. 5 indicates that the MLP model accurately predicted 2,514 true positives and 293 true negatives, with 179 false negatives and 1,014 false positives. This result indicates the model's ability to detect DED cases (high sensitivity) at the cost of more false alarms.

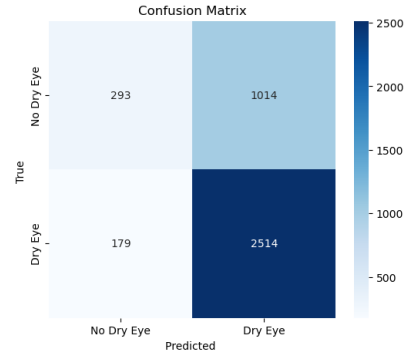


Fig. 5. Confusion Matrix

3) *Training Convergence*: The model trained in 20 epochs with dropout regularization, which prevented overfitting. Validation accuracy improved from 59.5% to 66.5% during training, with final holdout accuracy equal to the main test set performance.

4) *Visualization*:

- ROC Curve (Fig. 6): The ROC curve indicates an AUC of 0.57, which suggests limited discriminative ability between classes but supports the model's high recall.

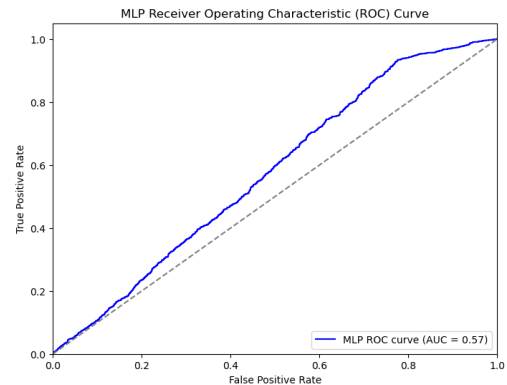


Fig. 6. ROC Curve

- Precision-Recall Curve (Fig. 7): The Area Under the PR Curve (AUPRC) was 0.7112, indicating strong perfor-

mance in identifying DED cases, especially under class imbalance.

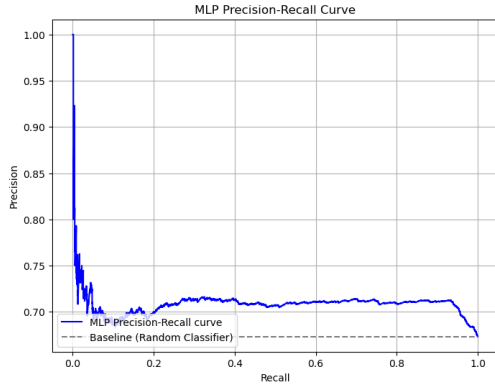


Fig. 7. Precision-Recall Curve

- Learning Curves (Fig. 8): The training and validation accuracy and loss curves indicated regular learning. Overfitting existed to a lesser degree, though the model produced consistent performance over validation data.

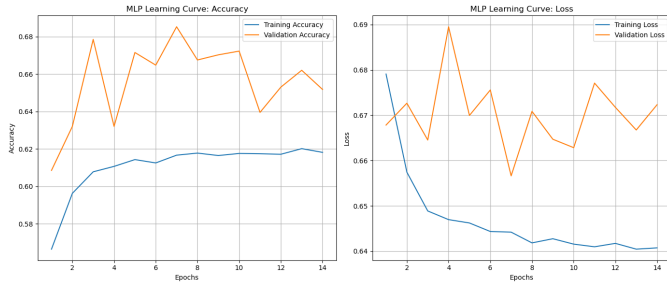


Fig. 8. Learning training and validation curves in accuracy (left) and loss (right)

Feature Correlation Heatmap

The heatmap Fig.9 indicates how closely the features are correlated with each other. Here:

- Most of the feature pairs have very low correlation values (close to 0).
- This implies the features are generally independent and don't overlap within the information that they offer.
- Independently, this is a good thing because it allows machine learning algorithms such as Logistic Regression and MLP to learn better without confusion from redundant or repetitive information.

In total, this proves that our chosen features are appropriate for constructing prediction models.

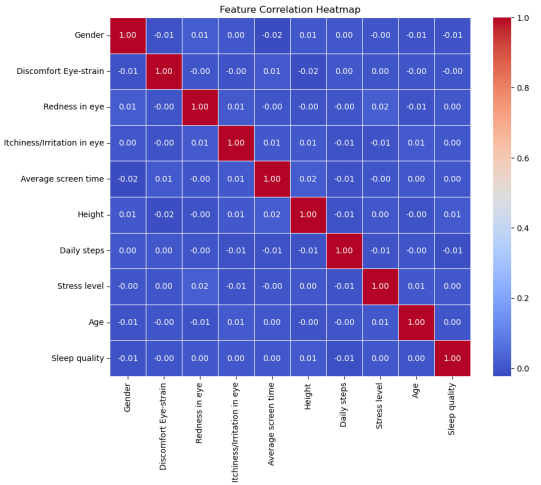


Fig. 9. Feature Correlation Heatmap

VI. COMPARATIVE ANALYSIS AND DISCUSSION

This section presents a simple comparison of the Logistic Regression (BLR) and Multi-Layer Perceptron (MLP) models along performance metrics, training behaviour, visual diagnostics, and real-world utility. Table III presents a side-by-side summary of significant quantitative results.

A. Quantitative Comparison

Comparison of Model Performance in Table III

TABLE III
PERFORMANCE COMPARISON: LOGISTIC REGRESSION VS MLP

Metric	Logistic Regression	MLP
Accuracy	0.65	0.69
Precision (DED class)	0.71	0.71
Recall (DED class)	0.81	0.92
F1-score (DED class)	0.76	0.80
ROC-AUC	0.58	0.58
Area under PR Curve (AUPRC)	0.71	0.71
CV Accuracy (5-fold)	55.4%	N/A
Epochs to Converge	N/A	< 20
RMSE (Predicted Probabilities)	0.4934	0.4800
RMSE (Adjusted Predictions)	0.5468	0.5461

B. Interpretation of Results

- Classification Performance
Both MLP and LR were on par for recall and F1-score for the DED class. MLP and LR both had good recall values (0.92 for MLP and 0.81 for LR), both indicating good detection ability for positive instances of DED, critical in clinical screening. Both of them, however, failed to detect instances of non-DED, with low recall value for class 0 for both MLP (0.22) and LR (0.22).
- ROC-AUC and AUPRC
ROC-AUC of both models was quite similar to one another (0.58 for LR and 0.57 for MLP), which means both models have the same threshold-independent discrimination ability. However, MLP has a higher AUPRC value (0.7067), which is especially useful for imbalanced

data sets like this one, so MLP is better suited for this specific task.

- Learning Behavior

The MLP showed smooth convergence within 20 epochs, showing successful training and generalization via mechanisms like dropout regularization. Logistic Regression, however, showed early overfitting, with a noticeable gap between training and validation losses, which means that the model will not generalize well across different subsets of data.

- Cross-Validation:

Logistic Regression's cross-validation accuracy was 55.4%, which suggests quite poor generalizability across different subsets of data. For the MLP, cross-validation accuracy was not provided, but its stable training behavior suggests greater robustness and generalization than LR.

- Interpretability and Real-World Use

Logistic Regression is highly interpretable, making it easier for clinicians to understand and use because it is a straightforward indication of the impact of individual features (like screen time and eye strain) on predictions. MLP is more powerful but a "black box" model and explainability is an important point in medical decision-making.

C. Limitations

- Class Imbalance:

Both models showed a bias towards the majority class (DED), leading to poor recall for the minority (non-DED) class. This imbalance limited the usefulness of accuracy as a performance metric, as it failed to reflect the models' capability to detect rare instances.

- Model Transparency:

The MLP model, while powerful, lacks interpretability—a crucial requirement in clinical settings where transparency and traceability of decisions are essential. In contrast, the Logistic Regression model is easier to interpret and therefore more aligned with healthcare applications.

- Shortage of Ensemble Methods:

Logistic regression model does not possess incorporated ensemble strategies or additional class-balancing techniques such as SMOTE (Synthetic Minority Over-sampling Technique) or advanced class weighting schemes. Incorporating these methods could enhance performance, particularly for the underrepresented non-DED class, and help mitigate the effects of class imbalance.

VII. CONCLUSION AND FUTURE WORK

This study compared two supervised algorithms—Logistic Regression (LR) and Multilayer Perceptron (MLP)—for classifying Dry Eye Disease (DED) based on lifestyle and symptom data. Both algorithms were compared on various performance measures including accuracy, precision, recall, F1-score, ROC-AUC, and AUPRC. The MLP model also outperformed LR on several key aspects, most noticeably with marginally better accuracy (0.69 vs. 0.65), DED class F1-score (0.80 vs. 0.76),

and AUPRC (0.71 vs. 0.67). MLP also achieved better training stability with minimal overfitting, indicating its superiority in dealing with class imbalance. It achieved a better recall (0.92 vs. 0.81) for DED cases, meaning MLP is slightly better in detecting DED-positive cases, a critical need in medical screening. Conversely, Logistic Regression was more transparent and interpretable, making it a more suitable choice for clinical settings where interpretability of feature contribution is paramount. While demonstrating reduced recall for the DED class, LR provides an easily interpretable, explainable model that allows clinicians to see how decisions are being made, a consideration for clinical adoption.

A. Conclusion

While MLP offers higher performance in the form of classification accuracy and curve-based metrics, this comes at the expense of lower interpretability. Logistic Regression, while being slightly weaker at the raw level of performance, is superior at areas of transparency and interpretability, which is paramount to clinician trust and clinical decision-making. Lastly, choice between these models must be based on the deployment context: MLP is better suited for machine-driven large-scale screening since it works better with imbalanced data, while Logistic Regression is better suited for clinician-assisted diagnosis, where model interpretability is paramount.

B. Future Work

To build on present findings and facilitate greater clinical utility, future research will involve:

- Integration of wearable sensor and mobile platform real-time patient data.
- Exploration of ensemble methods such as Random Forests and Gradient Boosting for improved predictive accuracy.
- Clinical validation through trials between model output and standard diagnostic procedure.
- Utilization of explainable AI (XAI) tools for enhancing the interpretability of complex models like MLPs.

These improvements aim to bridge the gap between model performance and clinical trust, accelerating AI adoption in ophthalmic screening and early intervention.

REFERENCES

- [1] S. Barabino, Y. Chen, S. Chauhan, R. Dana, and G. Van Setten, "Diagnosis and management of dry eye," *The Ocular Surface*, vol. 12, no. 4, pp. 191–204, 2014.
- [2] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, pp. 436–444, 2015.
- [3] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, et al., "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [4] J. Schmidhuber, "Deep learning in neural networks: An overview," *Neural Networks*, vol. 61, pp. 85–117, 2015.
- [5] A. Esteva, B. Kuprel, R. A. Novoa, J. Ko, S. M. Swetter, H. M. Blau, and S. Thrun, "Dermatologist-level classification of skin cancer with deep neural networks," *Nature*, vol. 542, no. 7639, pp. 115–118, 2017.